

README

Helios — Project Structure & Manifest

Repository restructured 2026-06-03 into `docs/{architecture,planning,strategy,archive}/`, `models/semantic/`, `eval/`, and (empty) implementation dirs (`dbt/` `mcp/` `agents/` `backend/` `frontend/` `tests/` `scripts/` `notebooks/`). The authoritative current layout is in `MIGRATION_REPORT.md` and `REPO_RESTRUCTURE_PLAN.md` . The detailed sections below predate the restructure — still accurate on *what Helios is*, but file paths have moved.

Paste this whole file into ChatGPT (or any LLM) to give it complete context on what has been built.

0. One-paragraph summary (read first)

Helios is an "Autonomous Growth Diagnosis Engine" — an always-on AI growth analyst that diagnoses **why** an e-commerce funnel moved, separates **mix-shift** (traffic composition changed) from **rate-change** (in-segment behavior changed), quantifies the movement as **revenue-at-risk in dollars**, prescribes a **prioritized, statistically-defensible experiment backlog**, and ships an executive **Decision Brief** — all grounded in **governed SQL it never hand-writes**, and graded by an **offline labeled benchmark**. It is deliberately **not** a dashboard and **not** a SQL chatbot; it runs autonomously. It is built on the public **Google Merchandise Store GA4 dataset** (`bigquery-public-data.ga4_obfuscated_sample_ecommerce` , Nov 2020–Jan 2021). Stack: **BigQuery + dbt + Python MCP servers (scipy/statsmodels/prophet) + Claude Agent SDK + Model Context Protocol + GitHub Actions CI**.

1. Current stage (important context)

This repo is currently a **complete specification + governed semantic layer + evaluation benchmark + interview prep — not yet running code**. Everything is *designed in depth*; implementation (the dbt models, the 5 MCP servers, the 7 agents, the eval harness) is fully planned in the docs but not yet written. So when reasoning about it, treat it as a **design/spec/benchmark project**, not a deployed system.

2. The architecture in 6 lines

- **5 MCP servers** — `warehouse-mcp` (the only BigQuery client; dry-run→run-query byte-budget gate; reconcile), `semantic-mcp` (the ONLY path to SQL; composes governed SQL from the metric registry → 0

hallucinated columns), `stats-mcp` (the ONLY path to math; the mix/rate decomposition, significance, forecasting), `experiment-mcp` (power/sample-size/test-design), `report-mcp` (Decision Brief + memory).

- **7 agents on a deterministic state machine** (the LLM composes tool calls, never controls flow):
Orchestrator → Monitor → Decompose → Diagnose → Critic → Prescribe → Narrator.
- **Semantic layer** — one governed YAML registry of every metric/dimension; the single source of truth.
- **Memory** — diagnosis history, suppression list, seasonality/launch calendars, action-tracking (did-the-fix-work loop).
- **Evaluation** — synthetic anomalies injected into a frozen GA4 copy; 50 labeled scenarios; target **≥85% root-cause accuracy vs ≤45% naive baseline**, 0 hallucinated columns, <5 min/run, ≤5 GiB/run.
- **Technical centerpiece** — mix-vs-rate decomposition: $\Delta R = \text{mix}(\sum \Delta w_i \cdot r_i) + \text{rate}(\sum w_i \cdot \Delta r_i) + \text{interaction}(\sum \Delta w_i \cdot \Delta r_i)$, which dissolves Simpson's paradox.

3. Folder tree

```
helios/
├── README.md · CLAUDE.md · .gitignore · MIGRATION_REPORT.md · REPO_RESTRUCTURE_PLAN.md
├── docs/
│   ├── architecture/ HELIOS_PROJECT_BIBLE · DATA_MODEL · DBT_GUIDE · MCP_ARCHITECTURE · AGENT_ARCHITECTURE · METRIC_DEFINITIONS
│   ├── planning/     DEPENDENCY_MAP · DEVELOPMENT_PLAN · IMPLEMENTATION_PLAYBOOK · LEAN_SCOPE
│   ├── strategy/     CLAUDE_CODE_WORKFLOW · INTERVIEW_GUIDE · RED_TEAM_REVIEW
│   └── archive/       (frozen build fragments: bible/interview/dbt-guide/data-model/playbook/red-team/claude-code-workflow)
├── models/semantic/ semantic_layer.yaml (PRODUCTION registry - 47 metrics, 19 dims, 7 entities)
├── eval/
│   ├── scenarios/    scenarios.yaml + 01-07-*.yaml + _VALIDATION.md (50-scenario benchmark)
│   └── benchmark_results/
├── dbt/ · mcp/ · agents/ · backend/ · frontend/ · tests/ · scripts/ · notebooks/ (implementation code - empty until ready)
```

4. File-by-file (what each is and why it matters)

Documentation (docs/)

File	What it is	Why it matters
HELIOS_PROJECT_BIBLE.md	The master spec — 25 sections: vision, business problem, personas, functional/non-functional requirements, data model, event model, funnel/revenue/attribution/metric definitions, semantic layer, analytics-engineering & dbt & BigQuery architecture, MCP architecture, agent architecture, memory, evaluation, experimentation, roadmap, interview & resume narratives.	The authoritative, rebuild-from-scratch reference. Everything else derives from it.
CLAUDE.md	Operating manual for any AI agent working on the repo: canonical metric/dimension/agent names, the grounding rules (G1–G5), dbt conventions, commands, the keystones that fail silently.	The continuity anchor — keeps every coding session consistent.

File	What it is	Why it matters
DEPENDENCY_MAP.md	Every artifact (T0–T13), what depends on what, the recommended generation order (milestones M0–M12), the keystones, and the critical path.	The build-time blueprint.
DEVELOPMENT_PLAN.md	The execution plan: work packages, the Claude Code workflow per activity, test/exit gates, a living milestone tracker, risk register.	How the project actually gets built, step by step.
MCP_ARCHITECTURE.md	Implementation spec for the 5 MCP servers: every tool's typed I/O, the guardrail chain, transport/auth, the error taxonomy, the per-agent tool allow-list, code skeletons.	Build spec for the tool layer.
AGENT_ARCHITECTURE.md	Implementation spec for the 7 agents: the deterministic FSM, the typed "Finding" hand-off envelope, the Critic refutation loop, the hypothesis-tree root-cause search, the revenue-at-risk formula, the tunable constants.	Build spec for the agent layer.
METRIC_DEPENDENCY_GRAPH.md	Dependency trees for every metric + RCA decomposition paths + the key identities (e.g. $\text{revenue_per_session} = \text{session_conversion_rate} \times \text{aov}$).	How metrics relate; drives root-cause analysis.
METRIC_GOVERNANCE_GUIDE.md	Governance for the semantic layer: the single-source-of-truth rule, the field-schema contract, versioning, how each agent consumes it, the validation/CI pipeline, the guardrail philosophy.	How the metric layer is owned and kept correct.
INTERVIEW_GUIDE.md	13 sections: project story, 5 elevator pitches, ~250 interview Q&A across Product/Growth/Data Analytics + Analytics Engineering + AI Engineering, system design, skeptic questions, resume bullets, STAR stories, founder round, top-1% prep — plus the "Helios Fact Sheet" cheat card.	How to present the project in interviews / placement.
sections/00–09...md	The 10 thematic source fragments the Bible was assembled from.	Source-of-truth fragments; the Bible is their concatenation.

Semantic layer (`models/semantic/`)

File	What it is	Why it matters
semantic_layer.yaml	The production registry. 47 metrics (34 named + 13 supporting) across revenue/traffic/funnel/conversion/retention/acquisition/product, 19 dimensions, 7 entities, 4 time grains. Each metric carries: business definition, SQL, aggregation, grain, supported dimensions, owner, caveats, common mistakes, guardrails (when-not-to-use / Simpson's-paradox / attribution /	The single source of truth for all metrics; no metric may exist outside it.

File	What it is	Why it matters
	segmentation), root-cause (upstream drivers, downstream impacts, decomposition path), which agents use it , validation checks, dbt/quality/anomaly/reconciliation tests , freshness. Referential-integrity validated (0 dangling references).	
<code>semantic_models.yml</code>	The v1 minimal registry (28 metrics, 16 dims).	Superseded by <code>semantic_layer.yml</code> ; kept for history.

Evaluation benchmark (`eval/scenarios/`)

File	What it is	Why it matters
<code>scenarios.yml</code>	The benchmark — 50 labeled e-commerce anomaly scenarios across 7 buckets (single/multi rate, single/multi mix, seasonality decoys, no-anomaly controls, data-quality artifacts). Each has the injection spec + 5 labels: anomaly, ground-truth, expected diagnosis, expected recommendation, expected revenue impact.	This is how Helios's accuracy is <i>proven</i> ($\geq 85\%$ vs $\leq 45\%$ baseline). The trust centerpiece.
<code>01_...07_*.yaml</code>	The 7 per-bucket source files.	Editable sources for the benchmark.
<code>_VALIDATION.md</code>	The verifier's report confirming canonical-name compliance, ID/coverage integrity, windows, revenue sanity.	Proof the benchmark is well-formed.

5. How the documents relate (the hierarchy)

```
HELIOS_PROJECT_BIBLE.md (master spec – the "why" and the full design)
├── derives → MCP_ARCHITECTURE.md (tool-layer build spec)
├── derives → AGENT_ARCHITECTURE.md (agent-layer build spec)
├── derives → semantic_layer.yml (the governed metrics) ── governed by → METRIC_GOVERNANCE_GUIDE.md
│                                                         └─ related by → METRIC_DEPENDENCY_GRAPH.md
├── graded by → eval/scenarios/scenarios.yml (the 50-scenario benchmark)
├── built via → DEPENDENCY_MAP.md + DEVELOPMENT_PLAN.md (what to build, in what order)
├── governed by → CLAUDE.md (operating rules for every session)
└── pitched via → INTERVIEW_GUIDE.md
```

6. Canonical vocabulary (so the LLM uses the right names)

- **Funnel:** `session_start` → `view_item` → `add_to_cart` → `begin_checkout` → `add_shipping_info` → `add_payment_info` → `purchase`
- **Session key** = `(user_pseudo_id, ga_session_id)` ; funnel flags are `reached_*` (max-downstream, monotonic).

- **10 channel groups:** Direct, Organic Search, Paid Search, Display, Paid Social, Organic Social, Email, Affiliates, Referral, Other.
- **Headline numbers:** $\geq 85\%$ root-cause accuracy vs $\leq 45\%$ naive baseline; 0 hallucinated columns; < 5 min/run; ≤ 5 GiB/run; 50 labeled scenarios; 47 metrics / 19 dimensions; ~ 250 interview questions; 25-section Bible.
- **The 3 trust pillars:** (1) governed-SQL grounding, (2) deterministic statistics, (3) adversarial verification + offline eval. Thesis: *the hard part isn't generating insights — it's making them correct and trusted*.
- **Honest limits:** the dataset is observational & ~ 3 months, so Helios *designs/sizes* experiments and runs quasi-experimental readbacks (not live A/Bs); LTV and cross-device identity are deferred; user metrics are cookie-grain.

7. Suggested intro line to paste before this file

"Here is the full structure and design of a portfolio project called **Helios**, an autonomous AI growth-diagnosis engine built on the GA4 Google Merchandise Store dataset. It is currently fully *specified* (architecture, governed semantic layer, and a 50-scenario evaluation benchmark) but not yet *implemented in code*. Read the manifest below, then [your question]."

Footnote: `CLAUDE.md` and the Bible currently sit under `docs/`; `CLAUDE.md` conventionally lives at the repo root so Claude Code auto-loads it (a trivial move). The `_build/` and `_interview_sections/` directories are intermediate scaffolding already merged into the final files and are safe to delete.